

Homework 3 File Submission

For both questions, assume A and B are sets with $A \subseteq B$.

** Prove does not mean give an example or show with a Venn diagram. I want a proof! **

A. Prove that $A \cup B = B$

Given: $A \subseteq B$

Let x be arbitrary, then:

$$\begin{aligned} A \cup B &= \{x \mid (x \in A) \vee (x \in B)\} \\ &= \{x \mid (x \in B) \vee (x \in B)\} \\ &= \{x \mid x \in B\} \\ &= B \end{aligned}$$

Definition of Union.

$A \subseteq B \rightarrow (x \in A \rightarrow x \in B)$. Definition of subset.

Idempotent Law on $\vee$.

Because x was arbitrary, $\forall x [x \in (A \cup B) \leftrightarrow x \in B]$.

$\therefore A \cup B = B$, given $A \subseteq B$. ■

B. Prove that $A \cap B = A$

Given: $A \subseteq B$

Let x be arbitrary, then:

$$\begin{aligned} A \cap B &= \{x \mid (x \in A) \wedge (x \in B)\} \\ &= \{x \mid (x \in A) \wedge (x \in A)\} \\ &= \{x \mid x \in A\} \\ &= A \end{aligned}$$

Definition of intersection.

$A \subseteq B \rightarrow (x \in A \rightarrow x \in B)$. Def. of subset.

Idempotent law on $\wedge$.

Because x was arbitrary, $\forall x [x \in (A \cap B) \leftrightarrow x \in A]$.

$\therefore A \cap B = A$, given $A \subseteq B$. ■